Teacher Guide

Lesson goal

Students should leave understanding that language models are next-token prediction systems whose behavior changes when training data changes.

Before-lesson checklist

- Python 3.11 or newer is available.
- The repository is cloned locally.
- Core dependencies are installed with `pip install -e .`.
- Optional Learn Mode dependencies are installed with `pip install -e ".[learn]"`.
- Sample files are present in `data/samples/`.
- The first training command has been tested once on the classroom machine.
- Projector or screen-sharing is ready for side-by-side output comparison.

Quick preflight:

```
python -m compileall src tests
python src/train.py --input_file data/samples/space_adventure.txt --out_dir
  runs/preflight --epochs 1 --batch_size 4 --seq_len 32 --d_model 64 --n_heads 4
  --n_layers 2 --device cpu
```

Quick-reference classroom flow

Section	Suggested timing	Teacher moves	Student outcome
1. Build it	10 min	Show byte tokens and ask for next-token predictions	Students form hypotheses
2. Train it	15 min	Run short training and track loss	Students connect loss to prediction error
3. Talk to it	10 min	Prompt and evaluate outputs	Students see uncertainty and repetition
4. Retrain it	15 min	Swap dataset and compare the same prompt	Students observe style shift
5. Reflect	10 min	Guide worksheet reflection	Students explain what changed and why

Suggested classroom pacing

45-minute version

1. Prediction warm-up (5 min)
2. Token viewer or short explanation (5 min)
3. Train the baseline model (12 min)
4. Generate and record output (8 min)
5. Retrain or show prepared comparison (10 min)
6. Exit ticket (5 min)

90-minute version

1. Prediction warm-up (10 min)
2. Tokenization and byte-token discussion (10 min)
3. Baseline training (15 min)
4. Prompt experiments (15 min)
5. Retrain comparison (20 min)
6. Attention and probability inspection (10 min)
7. Reflection and Q&A (10 min)

Facilitation tips

- Ask students to predict outputs before running commands.
- Reuse the exact same prompt before and after changing datasets.
- Project at least two outputs side-by-side for evidence-based discussion.
- Encourage precise uncertainty language: "I think", "I notice", "the output suggests".
- Normalize weird model behavior as expected in tiny models.
- Keep one prepared checkpoint ready in case live training is slow.

Common misconceptions and Q&A guidance

Misconception: "Lower loss means understanding"

Teacher response: Loss means prediction error went down; it does not prove understanding.

Misconception: "Attention shows thinking"

Teacher response: Attention is token weighting, not reasoning, consciousness, or intent.

Misconception: "If it sounds fluent, it must be true"

Teacher response: Fluency is pattern generation; factual claims still need external verification.

Misconception: "Retraining stores facts like a database"

Teacher response: Training shifts model weights so token patterns become more or less likely. It is not the same as saving a fact in a table.

Classroom Q&A prompts

- What changed between output before retrain and output after retrain?
- Which words or style markers suggest the dataset influenced behavior?
- Which parts of the output are fluent but not necessarily reliable?
- What does attention help us inspect?
- What does attention not prove?

Assessment ideas

Evidence	Emerging	Secure
Explains next-token prediction	Says the model "guesses words"	Connects previous tokens to next-token probabilities
Interprets loss	Says lower is better	Explains loss as prediction error, not understanding
Compares retrain outputs	Notices style changed	Uses vocabulary and tone evidence from both outputs
Explains attention	Says it shows important words	States attention is weighting, not consciousness

Troubleshooting quick guide

Training is too slow

Use fewer epochs, shorter sequence length, or the "Classroom demo" preset in Learn Mode.

Training text is too short

Use a longer sample file or lower `--seq_len`. The text must contain more tokens than the sequence length.

Outputs look repetitive

This is expected for tiny models and is a useful teaching moment.

CLI checkpoint load fails

Verify the training run saved `best.pt` or `last.pt` in the run directory.

Learn Mode experiment restore fails

Verify the experiment folder contains both `metadata.json` and `model.pt`.

Printable materials

Markdown sources live in `docs/`. Printable PDFs live in `docs/printable/`.

To regenerate the printables:

```
pip install -e ".[pdf]"
python tools/pdf/generate_printables.py
```

The default output is A4. For US Letter, run:

```
python tools/pdf/generate_printables.py letter
```

Reflection close-out (5–10 min)

Ask students to complete worksheet sections:

- Output Before Retrain
- Output After Retrain
- What Changed and Why?
- Reflection
- Extension exercises